



Threat Intelligence Integration Report

1. Introduction

Threat intelligence integration is a critical capability in modern Security Operations Center (SOC) environments that enhances threat detection and incident response by incorporating external intelligence data into internal security monitoring systems. Threat intelligence provides contextual information such as malicious IP addresses, file hashes, domains, and attacker behavior patterns that help security analysts identify and respond to cyber threats effectively.

In this task, threat intelligence feeds were integrated into **Wazuh SIEM**, and alerts were enriched using intelligence data from **AlienVault OTX**. The activity also included proactive threat hunting based on adversary techniques defined by the **MITRE ATT&CK framework**.

This exercise demonstrates how intelligence-driven security operations improve detection accuracy, reduce false positives, and enable proactive threat identification in SOC environments.

2. Objective

The objective of this task is to integrate threat intelligence into security monitoring workflows to enhance detection, enrichment, and threat hunting capabilities.

Key Objectives

- Import threat intelligence feeds into SIEM.
 - Enrich alerts using intelligence data.
 - Identify malicious indicators of compromise (IOCs).
 - Perform threat hunting using attacker techniques.
 - Improve incident detection accuracy using intelligence correlation.
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3. Tools Used

- Wazuh SIEM - Security monitoring and alert detection
 - AlienVault OTX - Threat intelligence feed and IOC validation
 - TheHive - Incident case management
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4. Methodology / Implementation

4.1 Threat Intelligence Feed Integration

Threat intelligence feeds containing indicators of compromise (IOCs) were imported into the Wazuh SIEM platform. These feeds include malicious IP addresses, file hashes, and known attacker infrastructure.

A test scenario was created using a mock suspicious IP address (192.168.110.140) to verify whether the SIEM system could match internal logs with external threat intelligence data.

Process

- Imported threat intelligence feed from AlienVault OTX.
- Configured Wazuh to automatically match network logs with IOC database.
- Triggered a test alert using a simulated malicious IP.
- Verified alert generation and enrichment.

Feed Integration Result

The SIEM successfully matched the suspicious IP with threat intelligence records and generated an alert indicating malicious activity.

4.2 Alert Enrichment Using Threat Intelligence

Alert enrichment involves adding contextual threat intelligence data to security alerts to improve investigation efficiency.

A Wazuh alert containing a suspicious IP address was enriched using intelligence from AlienVault OTX.

Alert Enrichment Table

Alert ID	IP Address	Reputation	Notes
003	192.168.110.140	Malicious	Linked to command-and-control infrastructure

Enrichment Observations

- The IP address was identified as part of known malicious infrastructure.
- Intelligence data provided risk classification and threat context.
- Enrichment helped prioritize the alert for immediate investigation.



4.3 Threat Hunting Using Intelligence

Threat hunting is a proactive security process where analysts search for hidden threats using intelligence-driven hypotheses.

A threat hunting activity was conducted to detect misuse of valid accounts mapped to MITRE ATT&CK technique **T1078 – Valid Accounts**.

Threat Hunting Query

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user.name != "system"
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This query identifies suspicious authentication activity involving non-system accounts.

Threat Hunting Summary

Threat hunting identified abnormal login activity from non-system user accounts accessing multiple systems within a short time period. The behavior deviated from normal baseline activity and suggested potential credential misuse. Intelligence mapping to attacker techniques enabled proactive detection and improved visibility into possible lateral movement attempts.

5. Indicators of Compromise (IOC) Analysis

Indicators of compromise are observable artifacts that indicate potential intrusion or malicious activity.

IOC Types Observed

IOC Type	Description	Example
Malicious IP	Attacker infrastructure	192.168.110.140
File Hash	Malware identifier	SHA256 hash
Suspicious Domain	Command-and-control domain	malicious-domain.com
User Behavior	Unauthorized login activity	Privilege misuse

Threat intelligence correlation confirmed malicious indicators and supported incident validation.

6. Findings

The threat intelligence integration process produced:



- Successful import of threat intelligence feeds.
- Automatic IOC matching within SIEM alerts.
- Improved alert context through enrichment.
- Detection of suspicious user account activity.
- Reduced false positives using intelligence validation.

Threat intelligence significantly improved investigation speed and detection accuracy.

7. Security Impact

Threat intelligence integration enhances SOC operations by:

- Providing real-time threat context.
 - Improving alert prioritization.
 - Enabling proactive threat detection.
 - Supporting faster incident response.
 - Enhancing organizational security posture.
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8. Conclusion

This task demonstrated the integration of threat intelligence into SOC monitoring workflows using Wazuh SIEM and AlienVault OTX. The exercise showed how intelligence feeds enhance alert enrichment, support threat hunting, and improve detection accuracy. Intelligence-driven security operations enable proactive defense mechanisms and strengthen incident response capabilities, making threat intelligence a critical component of modern cybersecurity operations.